

Madiha M. Shahid

AI/ML Engineer

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PROFESSIONAL SUMMARY

Final-year BS Computer Science student (KIET, CGPA 3.6/4.0) with 6+ months shipping AI/ML systems. Builds Generative AI apps around LLMs: prompt engineering, system/role prompts, Chain-of-Thought, structured JSON output, and output validation (Gemini, Vertex AI). Trains classical ML models and ships them in React, Django, and Streamlit. Expanding into RAG, embeddings, LangChain, and Hugging Face Transformers. Seeking intern or junior Generative AI / AI/ML roles.

EDUCATION

BS Computer Science — KIET University, Karachi | CGPA 3.6 / 4.0

Expected 2027

Intermediate (Computer Science) — NCR CET College, Karachi

2021 – 2022

Matriculation — The Educators, Karachi

2020

TECHNICAL SKILLS

Generative AI concepts: LLMs, Prompt Engineering, System / Role Prompts, Chain-of-Thought (CoT), Structured Output (JSON), Output Validation, RAG, Embeddings, LangChain, Hugging Face Transformers, Google Gemini API, Vertex AI

AI / Machine Learning: scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, OpenCV, Computer Vision, NLP, Feature Engineering, Model Evaluation

Programming Languages: Python, JavaScript, Java, C#

Web Development: React.js, Django, Node.js, Streamlit, REST APIs, HTML, CSS

Databases: MySQL, Microsoft SQL Server

Developer Tools: Git, GitHub, Docker, Jupyter Notebook, VS Code, Postman, Android Studio

WORK EXPERIENCE

Junior AI/ML Developer · Swag Kicks

Jan 2026 – Apr 2026

Karachi, Pakistan

- Owned the production Vertex AI chatbot: designed conversation flow, connected the LLM, and ran end-to-end tests before release.
- Trained and scored Vertex AI models, then retuned prompts and model settings from quality metrics until the bot was stable enough to deploy.
- Labeled computer-vision data with bounding boxes and led pre-launch QA: logged failure cases, fixed them, and signed off production readiness.

AI/ML Intern · Swag Kicks

Nov 2025 – Dec 2025

Karachi, Pakistan

- Built classification models in scikit-learn from raw tables: cleaning, feature extraction, train/test splits, and accuracy benchmarking.
- Wired model predictions into the product through React.js and Node.js REST APIs so the UI could show live ML results.

Machine Learning Intern · Elevvo

Sep 2025

Remote

- Ran the full ML loop on real datasets — EDA, cleaning, training, and hyperparameter tuning — to raise benchmark scores.
- Wrote reusable preprocessing pipelines for missing values, outliers, and categorical encoding so training data stayed consistent.

PROJECTS

Explain Like I'm a Bot React.js · Django · Gemini · Prompt Engineering

- Full-stack LLM app: pick a persona and topic; Gemini explains it; the browser reads it aloud (Intern, Professor, Journalist, Executive).
- System / role prompts: each persona has its own instruction (tone, length, no markdown) so one model speaks in four consistent voices.

Resume to Interview Questions Generator Streamlit · Gemini · Structured Output

- Gemini reads a resume and must return JSON only: name, skills, weak areas, 5–7 interview questions, and difficulty level.
- Structured output + validation: JSON-mode prompting plus a schema parser that drops hallucinated or broken replies before the UI.

Chain-of-Thought + Persona Reasoning Streamlit · Gemini · CoT Prompting

- Same problem, two Gemini calls: a short direct answer vs. a persona (e.g. Sherlock Holmes) that must reason step by step (Chain-of-Thought + role prompting) so both can be compared.

Jarvis – AI Voice Assistant Python · SpeechRecognition · pytsx3 · REST APIs

- Voice in → speech-to-text → NLP intent parse → tool call (weather/news REST APIs) → spoken reply; classmates used it as a daily assistant.

Face Recognition Attendance System Android Studio · FaceNet · ML Kit · Java

- On-device attendance: ML Kit detects the face, FaceNet embeddings identify the student, blink check reduces spoofing; stays reliable in mixed lighting.

Loan Approval Prediction System Python · scikit-learn · Pandas · NumPy

- Compared Logistic Regression, Decision Tree, and Random Forest with k-fold CV; RF won. Pipeline handles imputation and feature engineering end-to-end.

CERTIFICATIONS

- Google Cloud / Coursera — Introduction to AI and Machine Learning (Sep 2025)
- Simplilearn SkillUp — Deep Learning for Beginners (Jul 2026)
- Deloitte / Forage — Data Analytics Job Simulation (Jun 2026)
- HackerRank — Python (Basic) (Aug 2025)